

PhD Lecture in Behavioral Asset Pricing: Limits to Arbitrage

Frank Weikai Li

Associate Professor of Finance
City University of Hong Kong

- Behavioral finance applications to asset pricing posit that irrational investors affect prices
- There is a classic critique of this idea
 - The “arbitrage critique”
- According to this critique, irrational investors cannot affect prices for any significant amount of time
 - As soon as irrational investors move prices, this creates an attractive opportunity for rational investors
 - The rational investors trade against the mispricing, quickly eliminating it (“arbitrage”)
- A major achievement of behavioural finance is to successfully push back against the arbitrage critique
 - i.e. to show that there are “limits to arbitrage”

- **Terminology:**

- The “fundamental value” of an asset is its price in an economy with rational investors and no frictions
 - The price that properly reflects all available public information
 - The efficient market price
- In an economy with frictions, or where agents are not fully rational, an asset’s price can depart from fundamental value
 - This is a “mispricing”
 - Or an “inefficiency”
- Rational investors are sometimes referred to as “arbitrageurs”
- Less than fully rational investors are sometimes referred to as “noise traders”

- **Arbitrage**

- An investment strategy that guarantees a positive payoff in some state, with no possibility of a negative payoff, no net investment, and at arbitrary scale

- $\Rightarrow$ Money Machine

- Arbitrage opportunities are inconsistent with common notions of equilibrium

- If arbitrage opportunities are there, then someone will act, and prices will change, so not a stable situation

- **Law of one price**

- Two assets with the same payoff in every state must have the same price
- Key to law of one price is perfect substitutes (exact same payoffs)
- Underlies much of the design of derivatives

Limits to Arbitrage: Theory

- What is the response to the arbitrage critique?
 - The arbitrage critique says that it will be easy for rational investors to correct a mispricing
 - In reality, however, it is not easy
 - There are risks and costs that limit arbitrageurs' ability to correct a mispricing
 - This allows irrational investors to affect prices significantly and for a long time
 - Specific limits to arbitrage:
 - **Risks:** fundamental risk, noise trader risk, horizon risk, model risk
 - **Costs:** trading costs, shorting costs, identification costs

- **Fundamental risk**

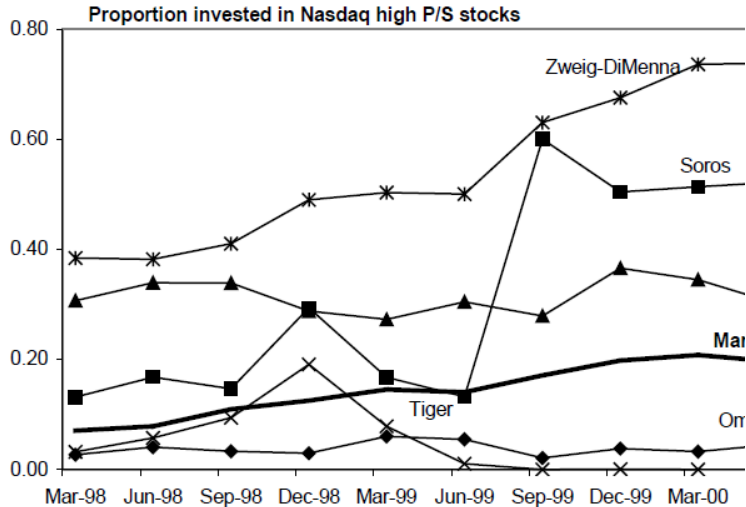
- The risk that there will be adverse news about the fundamental value of the mispriced asset
- Usually a problem because perfect substitutes are not available for what you really want to trade on
 - E.g., You think Ford is underpriced, but your best hedge against unexpected changes is GM

- **Noise trader risk** (De Long et al., 1990; Shleifer and Vishny, 1997)
 - The risk that, as a result of the mispricing worsening in the short run, the arbitrageur is forced to close out his trade prematurely at a loss
 - This risk arises because real-world arbitrageurs typically manage other people's money and because they use leverage
- **An example:**
 - Arbitrageurs engage in arbitrage in reaction to a stock mispricing
 - Mispricing persists for an extended period
 - Clients deem the arbitrageur incompetent and demand fund withdrawal
 - To deliver cash, manager must unwind the position at a loss
 - This threat causes managers to be less prone to take advantage of these opportunities, exacerbating the problem of pricing inefficiency

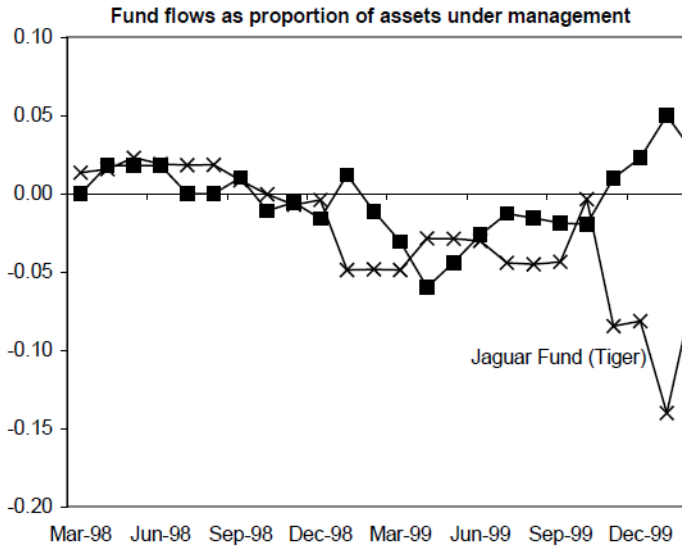
Limits to Arbitrage: Theory, ctd.

- Hedge funds' use of leverage and short selling compound the problem
 - If a fund borrows money to buy an undervalued asset and the mispricing then worsens
 - The lender, seeing the value of his collateral decline, may call the loan
 - This forces the fund to close out its trade at a loss
 - Similarly, if the fund shorts an overvalued asset and the asset then goes up in price
 - The lender will demand additional margin
 - If the fund cannot meet this demand, the short position will be closed out at a loss
 - A fund manager who foresees these potential outcomes will trade less aggressively ex-ante against the mispricing

Arbitrage In Reality



Arbitrage In Reality, ctd.



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- **Trading costs**

- Brokerage commissions
- Bid/Ask Spreads
- Price Impact

- **Shorting cost**

- Short selling involves borrowing shares from a lender, and need to pay lending fees
- The lending fees could be very high for certain stocks in certain periods
- Short selling are occasionally banned by regulators
- Some countries do not allow short selling at all

- **Information processing cost**

- Acquiring and processing financial statement data

Limits to Arbitrage: Theory, ctd.

- Thus far, we have examined the “static” version of the arbitrage critique
- There is also a dynamic version, which says:
 - Even if rational traders are unable to quickly correct a mispricing, assets will eventually become correctly priced
 - By trading sub-optimally, irrational traders become poorer over time
 - As a result, they eventually become irrelevant to the pricing of assets
- This idea has generated a sizeable literature on the “survival of noise traders”
- How compelling is the dynamic arbitrage critique?
 - It may take a very long time for irrational traders to exit market
 - New generation of irrational investors enter the market

Limits to Arbitrage: Evidence

- There are good theoretical reasons why arbitrage may be limited
 - But is there evidence that arbitrage is limited?
- Any example of a substantial, long-lived mispricing would be evidence of limits to arbitrage
 - But such examples are not easy to document
- Still, there are some pricing patterns that are widely viewed as mispricings
 - These constitute evidence of limits to arbitrage
 - And highlight the role played by each type of frictions
- Specifically, we look at:
 - Twin shares
 - Equity carve-outs
 - Index inclusions
 - Anomalies

Twin Shares

- Twin shares are distinct shares that are claims to the same cash flows
- In 1907, Royal Dutch and Shell merged their interests while remaining separate entities
 - Royal Dutch has a claim to 60% of the combined firm's cash flows
 - Shell has a claim to 40% of the combined firm's cash flows
- In an economy with rational investors and no frictions, the ratio of market values of equity should be:

$$\text{Price}_{RD} = 1.5 \times \text{Price}_S$$

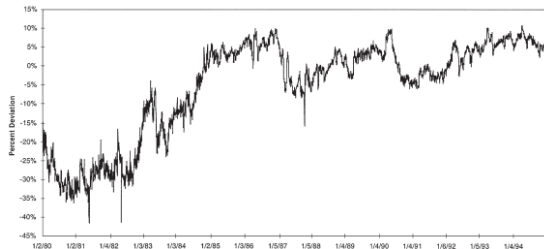
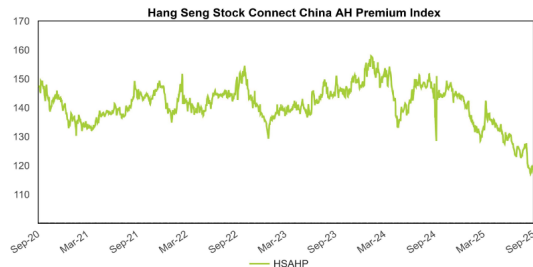


Fig. 1. Log deviations from Royal Dutch/Shell parity. Source: Froot and Dabora (1999).

This is a long-lived mispricing, and is therefore also evidence that it is hard to correct a mispricing

• Example: AH Premium

- Dozens of Chinese companies' stocks are listed in both China's domestic stock market (A shares) and Hong Kong stock market (H shares)
- A shares are usually traded at premium over H shares of the same company



Equity Carve-Outs

- Mitchell, Pulvino, and Stafford (2002) study equity carve-outs between 1985 and 2000
 - Cases where a parent company conducts an IPO for shares in a subsidiary firm and retains 80%+ ownership
- They find 70 cases where $MV_{Parent} < 0.8 \times MV_{Subsidiary}$
 - Implies that parents' "Stub" assets are negative
- Why is this mispricing hard to correct?
 - Fundamental risk (in 30% of cases, there is adverse fundamental news)
 - Noise trader risk
 - Shorting costs

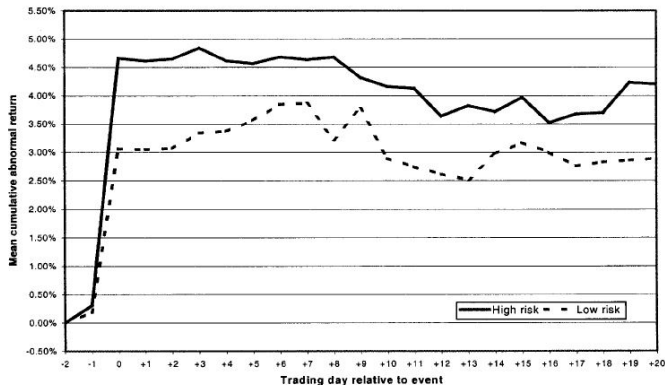
- **Palm/3Com**

- 3Com sells 5% of Palm in IPO, will spin off remainder in 9 months
- 1 Share of 3Com will own 1.5 Shares of Palm
- $P_{Palm} = \$95$
- P_{3Com} should be $\geq \$142$
- $P_{3Com} = \$81$
- Value of 3Com excluding Palm = $-\$60$!!
- Very few shares of Palm available to short

- Stocks are regularly added to the S&P 500 index to replace other stocks that have been removed due to merger or bankruptcy
- **Facts:** both on the announcement and execution days, the included stocks jump significantly in price by 3.5% on average, and the jump is not reversed
 - Harris and Gurel (1986), Shleifer (1986)
- The price jump is widely viewed as a mispricing
 - S&P's goal is to create a representative index, not to send signals about the included stocks' investment value
 - As such, the effect is evidence that there are limits to arbitrage
- To correct the mispricing, an arbitrageur would short the included stocks and buy substitute stocks
 - What are the limits to arbitrage in this case?

Index Inclusions, ctd.

- Wurgler and Zhuravskaya (2002) show that included stocks with worse substitutes (higher idiosyncratic risk) jump up more upon inclusion

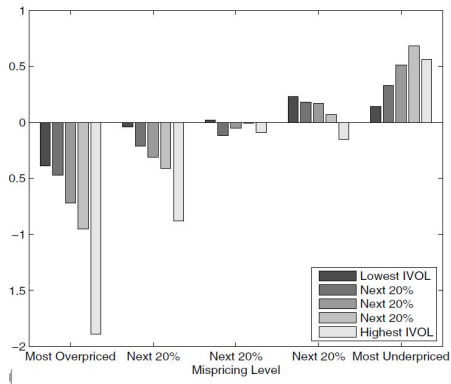


The Disappearing Index Effect

- The abnormal return associated with a stock being added to the S&P 500 has fallen from an average of 7.4% in the 1990s to less than 1% over the past decade (Greenwood and Sammon JF 2025)
- This has occurred despite a significant increase in the share of stock market assets linked to the index
- The main reason is professional active investors provide liquidity to accommodate the demand of passive indexers
- Although index trackers now buy about 7–8% upon index addition, total institutional ownership barely moves around index changes

Anomalies

- If anomalies are the result of mispricing, they should be more pronounced among stocks with more severe limits to arbitrage
 - Using institutional ownership as proxy for short-sale constraints, Nagel (2005) finds the short-leg of anomalies is stronger among stocks with lower institutional ownership
 - Using idiosyncratic volatility as proxy for arbitrage risk, Stambaugh, Yu and Yuan (2015) find mispricing is amplified among stocks with highest idiosyncratic volatility



Stambaugh, Yu and Yuan (2015)

The Causal Effects of Limits to Arbitrage

- Chu, Hirshleifer and Ma (2020) show the causal effects of limits to arbitrage on asset pricing anomalies
- They use Regulation SHO as a quasi-experiment:
 - SEC removes short sale restrictions on a randomly selected group of stocks (pilot stocks) within Russell 3000 index
 - The pilot program period is from May 2, 2005 to August 6, 2007
- Research design is standard differences-in-differences test:
 - Construct anomaly long-short portfolios for pilot stocks and non-pilot stocks separately
 - Compare the return spread of the long-short portfolios for pilot firms relative to non-pilot firms, before and during the pilot period
- Their main findings are:
 - Anomalies become weaker for pilot stocks relative to non-pilot stocks during the pilot period
 - The effect comes mostly from the short leg portfolio

The Causal Effects of Limits to Arbitrage, ctd.

The main difference-in-differences test employs the following specification:

$$r_{it} = \alpha_t + \beta Pilot_i \times During_t + \beta_1 Pilot_i + \epsilon_{it}, \quad (1)$$

where r_{it} is the monthly return of portfolio i , which can be the long leg, the short leg, or the long-short portfolio of an anomaly, in month t ; α_t is the time fixed effects; $Pilot_i$ is a dummy variable which is equal to one if portfolio i is formed on pilot firms, and zero otherwise; $During_t$ is a dummy variable, which equals one if month t is between June 2005 and July 2007, i.e. when the pilot program of Regulation SHO takes action. Since $During_t$

Chu, Hirshleifer, and Ma (2020)

Information Processing Costs as Limit to Arbitrage

- Bowles, Reed, Ringgenberg and Thornock (JF 2024)
- Anomaly returns are concentrated in the first month after anomaly-relevant information is first released, and the returns decay quickly thereafter

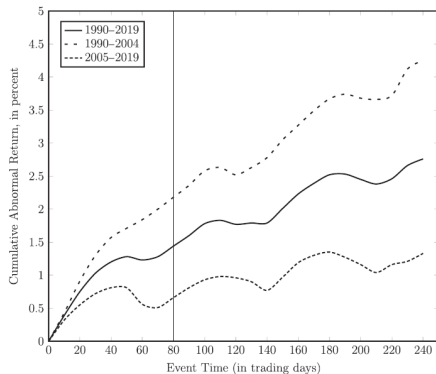


Figure 1. Anomaly returns in event time. The figure shows the profile of abnormal returns for the average anomaly (equally weighted) in event time. The figure shows the returns across the entire sample (1990 to 2019), across the early period of our sample (1990 to 2004), and across the recent years of our sample (2005 to 2019). The solid vertical bar at $t = 80$ trading days indicates when the FF92 rebalancing date occurs, on average.

- The literature on limits to arbitrage has been influential
 - There is now wide agreement among academics (and practitioners) that arbitrage is limited
 - Albeit some disagreement as to how limited it is
- This success was one reason why behavioural finance “took off” in the 1990s
- Still, we should not be complacent
 - Whenever we argue that irrational investors’ demand affect prices, we should ask if arbitrage is truly limited